

Create RDS instance in AWS:

hamariyusql

Summary

DB identifier

hamariyusql

CPU

3.13%

Status

Available

Class

db.t4g.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

us-east-1c

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations - new

Tags

Recommendations

Connectivity & security

Endpoint & port

Endpoint

hamariyusql.cgqj3fbcprx.us-east-1.rds.amazonaws.com

Port

3306

Networking

Availability Zone

us-east-1c

VPC

vpc-0e3aab9e0a5d6aaa0

Subnet group

rds-ec2-db-subnet-group-1

Subnets

subnet-00e0f93a02e0bc9b0  
subnet-0999622ec41339d3e  
subnet-0e8da272f32fe980f  
subnet-09ad130d03487413f  
subnet-0e90fb6754e558fae

Network type

IPv4

Security

VPC security groups

rds-ec2-2 (sg-0b60317375475dbfe)

Active

Publicly accessible

No

Certificate authority

Info

rds-ca-rsa2048-g1

Certificate authority date

May 26, 2061, 01:34 (UTC+02:00)

DB instance certificate expiration date

January 06, 2026, 21:13 (UTC+01:00)

Creation of EMT instance with bundled applications such as Hadoop, HBase , Sqoop.

Amazon EMR > EMR on EC2: Clusters > HamariyuCluster

Updated less than a minute ago

Terminate

Clone in AWS CLI

Clone

HamariyuCluster

Summary

Cluster info

Cluster ID

j-3N2FFXW2F865H

Cluster configuration

Instance groups

Capacity

1 Primary | 1 Core | 1 Task

Applications

Amazon EMR version

emr-5.30.1

Installed applications

HBase 1.4.13, Hadoop 2.8.5, Hive 2.3.6, Hue 4.6.0, Pig 0.17.0, Sqoop 1.4.7

Cluster management

Log destination in Amazon S3

aws-logs-286523281873-us-east-1/elasticmapreduce

Primary node public DNS

ec2-44-211-249-229.compute-1.amazonaws.com

Connect to the Primary node using SSH

Connect to the Primary node using SSM

Status and time

Status

Starting

Creation time

January 06, 2025, 21:13 (UTC+01:00)

Elapsed time

13 minutes, 9 seconds

Properties

Bootstrap actions

Instances (Hardware)

Steps

Applications

Configurations

Monitoring

Events

Tags (0)

Cluster logs

Archive log files to Amazon S3

Turned on

Encryption for logs

Turned off

Cluster termination and node replacement

Termination option

Automatically terminate cluster after idle time

Idle time

1 hour

Termination protection

Off

Unhealthy node replacement

On

Network and security

Network

Virtual Private Cloud (VPC)

vpc-0e3aab9e0a5d6aaa0

Subnet(s) and Availability Zone(s) (AZ)

subnet-0a90f090ad750965a us-east-1b

EC2 security groups (firewall)

Security configuration

Security configuration

None

EC2 key pair

HAMARIYU

Permissions

Service role for Amazon EMR

EMR\_DefaultRole

EC2 instance profile

EMR\_EC2\_DefaultRole

Custom automatic scaling role

Connect RDS with EMR instance .Set up an EC2 connection with RDS:

RDS > Databases > Set up EC2 connection

Step 1: Set up EC2 connection

Select EC2 instance

Database

hamariyusql

EC2 instance

i-0d09acfbfaaf35d78

Create EC2 instance

Connection already exists

A connection between RDS database hamariyusql (security group rds-ec2-2) and EC2 instance i-0d09acfbfaaf35d78 (security group ec2-rds-2) already exists.

Cancel

Continue

Login to RDS through EMR instance using command

```

login as: hadoop
Authenticating with public key "imported-openssh-key"
Last login: Mon Jan 6 20:16:52 2025 from 77-172-173-96.fixed.kpn.net

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      _ | ( _ _ | /
      _ | \ _ _ | _ |
      _ | \ _ _ | _ |

Amazon Linux 2 AMI

https://aws.amazon.com/amazon-linux-2/
95 package(s) needed for security, out of 160 available
Run "sudo yum update" to apply all updates.

EEEEEEEEEEEEEEEEEEEE MMMMMMM      MMMMMMM RRRRRRRRRRRRRR
E::::::::::::::::::::E M::::::::M      M::::::::M R:::::::::R
EE::::::::EEEEEEEE::E M::::::::M      M::::::::M R:::RRRRRR:::R
  E:::E      EEEEE M::::::::M      M::::::::M RR:::R      R:::R
  E:::E      M:::M M:::M M:::M M:::M R:::R      R:::R
  E:::EEEEEEEEEE M:::M M:::M M:::M M:::M R:::RRRRRR:::R
  E::::::::::::E M:::M M:::M M:::M M:::M R:::::::::RR
  E:::EEEEEEEEEE M:::M M:::M M:::M R:::RRRRRR:::R
  E:::E      M:::M M:::M M:::M R:::R      R:::R
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EEEEEEEEEEEEEEEEEEEE MMMMMMM      MMMMMMM RRRRRRR      RRRRRR

[hadoop@ip-172-31-84-218 ~]$ mysql -h hamariyusql.cgqj3fbcprxr.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
^C
[hadoop@ip-172-31-84-218 ~]$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 59
Server version: 5.5.68-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>

```

```
create database taxi_records
```

(VendorID INT, tpep\_pickup\_datetime VARCHAR(50), tpep\_dropoff\_datetime VARCHAR(50), Passenger\_count INT, Trip\_distance FLOAT, RatecodeID INT, store\_and\_fwd\_flag VARCHAR(2), PULocationID INT, DOLocationID INT, payment\_type INT, fare\_amount FLOAT, extra FLOAT, mta\_tax FLOAT, tip\_amount FLOAT, tolls\_amount FLOAT, improvement\_surcharge FLOAT, total\_amount FLOAT, congestion\_surcharge FLOAT, Airport\_fee FLOAT );

```
[hadoop@ip-172-31-84-218 ~]$ mysql -h hamariyusql.cgqj3fbcprxr.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
^C
[hadoop@ip-172-31-84-218 ~]$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 59
Server version: 5.5.68-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database taxi records
-> ;
Query OK, 1 row affected (0.00 sec)

MariaDB [(none)]> CREATE TABLE taxi_records.trip_log
-> (VendorID INT, tpep_pickup_datetime VARCHAR(50), tpep_dropoff_datetime VARCHAR(50),
-> Passenger_count INT, Trip_distance FLOAT, RatecodeID INT, store_and_fwd_flag VARCHAR(2),
-> PULocationID INT, DOLocationID INT, payment_type INT, fare_amount FLOAT, extra FLOAT,
-> mta_tax FLOAT, tip_amount FLOAT, tolls_amount FLOAT, improvement_surcharge FLOAT,
-> total_amount FLOAT, congestion_surcharge FLOAT, Airport_fee FLOAT );
Query OK, 0 rows affected (0.00 sec)

MariaDB [(none)]> Ctrl-C -- exit!
Aborted
[hadoop@ip-172-31-84-218 ~]$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 65
Server version: 5.5.68-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Download required csv files from internet into local using the below command

wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow\_tripdata\_2017-01.csv

wget https://nyc-tlc-upgrad.s3.amazonaws.com/yellow\_tripdata\_2017-02.csv

```
[hadoop@ip-172-31-84-218 ~]$ curl https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
--2018-01-06 20:41:11-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-01.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 54.231.192.1, 54.231.164.201, 54.9.145, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|54.231.192.1|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 914029540 (872M) [text/csv]
Saving to: 'yellow_tripdata_2017-01.csv'

100K[=====] 914,029,540 52.4MB/s   in 17s

2018-01-06 20:41:28 (51.1 MB/s) - 'yellow_tripdata_2017-01.csv' saved [914029540/914029540]

[hadoop@ip-172-31-84-218 ~]$ curl https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
--2018-01-06 20:41:34-- https://nyc-tlc-upgrad.s3.amazonaws.com/yellow_tripdata_2017-02.csv
Resolving nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)... 52.217.232.241, 54.231.228.113, 54.24.205, ...
Connecting to nyc-tlc-upgrad.s3.amazonaws.com (nyc-tlc-upgrad.s3.amazonaws.com)|52.217.232.241|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 863487050 (823M) [text/csv]
Saving to: 'yellow_tripdata_2017-02.csv'

100K[=====] 863,487,050 57.1MB/s   in 15s

2018-01-06 20:41:49 (54.6 MB/s) - 'yellow_tripdata_2017-02.csv' saved [863487050/863487050]

[hadoop@ip-172-31-84-218 ~]$
```

To load data into mysql table and run the below command

LOAD DATA LOCAL INFILE '/home/hadoop/yellow\_tripdata\_2017-01.csv' INTO TABLE taxi\_records.trip\_log FIELDS TERMINATED BY ',' IGNORE 1 LINES;

LOAD DATA LOCAL INFILE '/home/hadoop/yellow\_tripdata\_2017-02.csv' INTO TABLE taxi\_records.trip\_log FIELDS TERMINATED BY ',' IGNORE 1 LINES;

```
2025-01-06 20:41:49 (54.6 MB/s) - 'yellow_tripdata_2017-02.csv' saved [863487050/863487050]

[hadoop@ip-172-31-84-218 ~]$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 66
Server version: 5.5.68-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> LOAD DATA LOCAL INFILE '/home/hadoop/yellow_tripdata_2017-01.csv' INTO TABLE taxi_records.trip_log FIELDS TERMINATED BY ',' IGNORE 1 LINES;
Query OK, 9710820 rows affected, 65535 warnings (1 min 15.20 sec)
Records: 9710820 Deleted: 0 Skipped: 0 Warnings: 19421640

MariaDB [(none)]>
MariaDB [(none)]> LOAD DATA LOCAL INFILE '/home/hadoop/yellow_tripdata_2017-02.csv' INTO TABLE taxi_records.trip_log FIELDS TERMINATED BY ',' IGNORE 1 LINES;
Query OK, 9169775 rows affected, 65535 warnings (1 min 9.82 sec)
Records: 9169775 Deleted: 0 Skipped: 0 Warnings: 18339550

MariaDB [(none)]> █
```

```
MariaDB [(none)]> select count(*) from taxi_records.trip_log;
+-----+
| count(*) |
+-----+
| 18880595 |
+-----+
1 row in set (9.27 sec)

MariaDB [(none)]> █
```